Semi-Supervised Multichannel Anomaly Detection Using QAOA-Driven Selection and Set Covering

Anonymous Authors

***Abstract*—Anomaly detection in multivariate time series remains a challenging task due to high dimensionality, temporal dependencies, and complex inter-variable relationships. In this work, we extend a previously proposed hybrid quantum-classical anomaly detection framework to the multivariate setting. The method formulates anomaly detection as a Quadratic Unconstrained Binary Optimization problem solved via the Quantum Approximate Optimization Algorithm, where multichannel information is implicitly encoded in a problem-dependent cost matrix. The quantum optimization stage yields multiple high-quality candidate solutions, which are subsequently exploited with a semi-supervised fashion approach by a custom set-covering strategy, with the goal to construct interpretable regions of nominal behavior in the joint time-value domain. The proposed method is validated on a unpublished real-world dataset of water smart meter measurements, demonstrating robust and competitive detection performance. The goal is to also enable empirical analysis of solution multiplicity and near-degeneracy effects in the quantum optimization landscape, along with their impact on the induced geometric representations of normality.**

I. Introduction

In modern technological systems, such as industrial monitoring platforms, cyber-physical infrastructures, and large-scale sensing networks, the continuous monitoring of critical variables has become increasingly essential [1]–[3]. The correct operation of these systems, commonly referred to as expected behavior, can be compromised by the occurrence of anomalies, i.e., rare and unpredictable events that may signal faults, degradations, or abnormal operating conditions [4]. As systems grow in complexity and scale, anomaly detection (AD) becomes progressively more challenging, calling for inference strategies capable of dealing with high dimensionality, non-stationarity, and intricate dependency structures [5].

These challenges are further amplified in multivariate time series scenarios, where system dynamics cannot be accurately described by a single observable but instead arise from the joint evolution of multiple correlated variables, making anomalies harder to detect and interpret [6]. Moreover, in real-world applications a small amount of anomaly-related information is often available, while exhaustive labeling remains impractical. This motivates semi-supervised AD strategies that exploit weak supervision to improve robustness and interpretability without relying on large annotated datasets [7]. In this context, quantum-inspired and quantum-assisted optimization methods have recently emerged as promising alternatives for tackling high-dimensional combinatorial problems underlying AD, particularly when global consistency constraints and non-convex cost landscapes are involved [8], [9]. Variational Quantum Algorithms (VQAs) [10] offer a natural framework for tackling the structure of such optimization problems from a different perspective, potentially reaching competitive detection performance with classical state-of-the-art paradigms, and enabling richer modeling of normal system behavior beyond what is accessible to classical solvers alone [11].

Building upon our previous work [12], in this paper we propose a hybrid quantum–classical framework for multivariate time series AD. In particular, we consider a multivariate time series as an ordered collection of observations acquired over time from multiple correlated channels:

$$\mathbf{X} = \left\{ \mathbf{x}_t \in \mathbb{R}^C \mid t = 1 \dots T \right\}, \tag{1}$$

where T denotes the number of time steps (observations), C is the number of channels, and $\mathbf{x}_t = (x_t^{(1)}, x_t^{(2)}, \dots, x_t^{(C)})$ represents a vector of measurements collected at time t.

Our proposal leverages the Quantum Approximate Optimization Algorithm (QAOA) [13] to solve a Quadratic Unconstrained Binary Optimization (QUBO) problem equivalent to the AD task of interest [14], in which the multivariate data structure is implicitly encoded. The solutions produced by the quantum optimization stage are then exploited by a dedicated set-covering algorithm, which interprets the selected samples as representative centers of normal behavior and constructs convex-hulls as confidence regions in the joint time-value domain, ultimately defining the predicted nominal behavior.

Experimental results demonstrate that the proposed approach achieves high detection performance under realistic operating conditions, characterized by strong class imbalance and localized anomalies. In addition to competitive numerical performance with respect to classical baselines, empirical analyses highlight distinctive properties of the quantum optimization stage, including the availability of multiple meaningful solutions in parallel and the management of otherwise ambiguous scenarios related to multiple optima.

II. Problem Formulation and Main Background

We begin by introducing the datasets considered in this work and the challenges they pose, then briefly recall the core formulation, notation, and optimization pipeline of our previous univariate approach, appropriately adapted for the multivariate case.

A. Dataset

Within the multivariate setting defined in (1), an anomaly is defined at the timestamp level: a time index t is labeled as anomalous if at least a component of $\mathbf{x}_t \in \mathbf{X}$ is an anomaly (in the univariate sense). Having that clarified, the datasets

considered in this study are characterized by the presence of localized and rare anomalies, accounting for at most 5-6% of the total observations, resulting in a strongly imbalanced setting that reflects typical real-world monitoring scenarios. This choice clearly distinguishes the AD problem addressed in this work from pattern recognition-oriented formulations, where anomalies often correspond to extended abnormal sequences or recurring patterns rather than sparse events.

The methodological development and preliminary validation of our work were carried out using synthetic multivariate datasets, generated by independently sampling nominal observations for each channel from a selected probability distribution (e.g., uniform or exponential), while anomalous samples are drawn from a different distribution with altered parameters. Normal and anomalous samples are then concatenated and randomly shuffled, yielding multivariate time series with a controlled contamination level.

The main contribution of this work is based on a real-world dataset addressing a highly challenging issue with significant economic and societal implications. It consists of aggregated water consumption measurements (liters/hour) collected during the 2024-2025 period from residential facilities deployed within a water distribution network in {`site name anonymized for review`}. Measurements are organized at the level of hydraulic districts, which reflect the underlying physical topology of the distribution network. For each district, the aggregated consumption of more than 200,000 facilities defines a multivariate time series whose channels are temporally aligned and intrinsically correlated due to shared infrastructure and usage patterns. As a consequence, both nominal behavior and anomalous events (e.g., leaks) are expected to manifest coherently across neighboring channels.

To set up the problem, preprocessing is involved in the steps of normalization, data cleaning for corrupted measurements and highly irregular usage profiles, and anomaly injection. The latter is designed to emulate sudden water losses or irregular drops in consumption intended to simulate fraudulent behaviors such as meter tampering or manipulation. An overview of the datasets is reported in Table I, where γ is the contamination level and ϵ represents the percentage of discarded observations.

TABLE I
DATASET SUMMARY

District	C	**Data type**	**Observ.**	γ	ϵ
artificial$_2$	2	numerical	10.0k	5%	0%
artificial$_3$	3	numerical	10.0k	5%	0%
artificial$_4$	4	numerical	10.0k	6%	0%
`site anonym.`	4	numerical	7.8k	6%	4%
`site anonym.`	3	numerical	8.2k	6%	2%
`site anonym.`	4	numerical	7.7k	6%	4%
`site anonym.`	2	numerical	6.0k	5%	6%

{`Names of geographical sites anonymized for review`}

B. Adopted Notation and Proposed Method

Given a univariate time series $\{x_i\}_{i=1}^T$, each observation can be associated with a binary decision variable $z_i \in \{0,1\}$, indicating whether the corresponding sample is selected ($z_i = 1$) or not ($z_i = 0$). The entire dataset is thus embedded into a binary vector $\mathbf{z} \in \{0,1\}^T$, enabling a combinatorial representation of the selection problem as:

$$f_\Theta(\mathbf{z}) = \alpha \sum_i \delta_\phi(x_i)\, z_i \;+\; \alpha' \sum_{i<j} d(x_i, x_j)\, z_i z_j\,. \quad (2)$$

This representation encodes information about the structure of the data through two complementary contributions: $\delta_\phi(x_i)$ measures the deviation of sample x_i from a reference statistical model ϕ fitted to the time series, while $d(x_i, x_j)$ quantifies the pairwise dissimilarity between samples, balanced by tuning the parameters parameters $\Theta = \{\alpha, \alpha'\}$. Clearly, (2) admits the trivial solution $\mathbf{z} = \mathbf{0}$ as a global minimizer, but as discussed in [12], this issue can be mitigated by incorporating suitable constraints or rewards that favor non-degenerate selections.

The function in (2) defines a QUBO problem $\mathcal{Q}(z) = z^\top \mathbf{M}\, z$, where $\mathbf{M} = (M_{ij})$ is symmetric and such that:

$$\mathbf{M} = \begin{pmatrix} \delta_\phi(x_1) & d(x_1,x_2) & \dots & d(x_1,x_T) \\ d(x_2,x_1) & \delta_\phi(x_2) & \dots & d(x_2,x_T) \\ \vdots & \ddots & & \vdots \\ \vdots & & \ddots & \vdots \\ d(x_T,x_1) & d(x_T,x_2) & \dots & \delta_\phi(x_T) \end{pmatrix} \quad (3)$$

In order to solve the resulting combinatorial problem with a VQA, such as QAOA, the binary variables are mapped onto spin variables $\sigma_i \in \{\pm 1\}$ via $z_i = (1-\sigma_i)/2$ and then promoted to Pauli$-z$ gates $\boldsymbol{\sigma}_z^{(\cdot)}$ yielding the quantum observable

$$H_f(\boldsymbol{\sigma}) = \sum_i M'_{ii} \boldsymbol{\sigma}_z^{(i)} + \sum_{i,j} M'_{ij} \boldsymbol{\sigma}_z^{(i)} \boldsymbol{\sigma}_z^{(j)}\,, \quad (4)$$

where M'_{ij} for $i \neq j$ is derived from $d(x_i, x_j)$ in (3) and M'_{ii} is derived from $\delta_\phi(x_i)$ in (3), with suitable normalizations to ensure that (4) is indeed an observable. At this point, following the standard QAOA optimization routine, the quantum circuit associated to H_f, produces a probability distribution over binary strings. High-probability strings correspond to near-optimal solutions of the original QUBO problem, providing candidate subsets of selected samples [15] that we can interpret as representative pivots of nominal system behavior, serving as anchors for a subsequent geometric post-processing stage.

C. QUBO Formulation for Multivariate Anomaly Detection

In the multivariate setting, the core differences arise in the structure of the QUBO cost matrix, which is modified to jointly encode information from multiple channels. Considering datasets as in (1), each channel induces a univariate time series, for which the cost construction in (2) can be independently applied. This simple generalization is insufficient since the corresponding independent blocks cannot encode relationships between different channels. For this reason, we must operate on vector-valued observations.

Specifically, the univariate distance term $d(x_i, x_j)$ in (2) should be replaced by any correlation-aware

multivariate distance defined on $\mathbb{R}^C$; for instance, we can adopt the squared Mahalanobis distance $d_M(\mathbf{x}_i, \mathbf{x}_j) = (\mathbf{x}_i - \mathbf{x}_j)^\top \boldsymbol{\eta}^{-1}(\mathbf{x}_i - \mathbf{x}_j)$, where $\boldsymbol{\eta}$ denotes the empirical covariance matrix estimated from data. Similarly, the statistical deviation term is defined with respect to a global multivariate reference model ϕ fitted to the joint distribution of the data, such as a multivariate moving-average baseline. As a result, both diagonal and off-diagonal contributions of the QUBO cost encode genuinely multivariate information. The decision variables yielded by QAOA are indexed by time, and the solution $\mathbf{z} = z_1 \dots z_T \in \{0,1\}^T$ defines a subset of timestamps whose associated vectors $\{\mathbf{x}_t \mid z_t = 1\}$ are selected as representative samples of nominal behavior.

Another possibility of extension involves the encoding of multivariate dependencies via inter-channel coupling submatrix blocks. In this case the QUBO matrix has the form $\mathbf{M} = \mathbf{M}_0 + \lambda \mathbf{M}_p$, where $\mathbf{M}_p$ contains off-diagonal coupling blocks that encode interactions between different channels, while $\mathbf{M}_0$ is a block-diagonal matrix whose blocks are the matrices M_c constructed independently for each individual channel as in (3). Such couplings may be defined using empirical inter-channel correlations, learned dependency structures, or other domain-specific similarity measures.

While the last formulation preserves a modular channel-wise structure, it requires explicit design choices for the coupling terms and introduces additional hyperparameters; furthermore, it represents a partial distortion of the original problem, leveraging scalar observations by concatenating channel-wise time series, leading to a QUBO problem in which each decision variable selects individual channel samples rather than entire multivariate observations. For this reason, the fully multivariate formulation described above is adopted as the primary approach.

D. Set-covering Strategy

As discussed, in the multivariate setting, each timestamp t is associated with an observation vector $\mathbf{x}_t \in \mathbb{R}^C$. In order to jointly encode the temporal information, we here work in the augmented time-value space $\mathbb{R}^{C+1}$. In such space, each candidate solution produced by QAOA identifies a subset that (approximately) minimize the objective function. We can thus define anchors for each selected index t_i:

$$\mathbf{b}_i = (t_i, \mathbf{x}_{t_i}) \in \mathbb{R}^{C+1}\,, \tag{5}$$

which provide representative starting points around which local regions of normality are constructed via set-covering: starting from $\mathbf{b}_i$, an expansion parameter $k \in \mathbb{N}$ and a neighborhood

$$\mathcal{N}_k(\mathbf{b}_i) = \{(t, \mathbf{x}_t) \mid |t - t_i| \le k\}\,, \tag{6}$$

we progressively collect normal samples from the training time series (i.e., all normal points within k time-steps from t_i) and build their convex hull. The expansion is halted at the smallest $k^\star$ such that including $\mathcal{N}_{k^\star+1}(\mathbf{b}_i)$ would introduce at least one anomalous sample according to the labels available in the training set. The halting occurs also when the maximum expansion $k_{\max}$ is reached. For the fashion in which the learning process is constructed and carried out, our architecture falls into the family of semi-supervised methods (despite that, it is possible to define a mechanism for creating such polygons without the need for labels, similarly to [16]).

To further elaborate specifically on this reasoning, let $\{\mathbf{v}_{i,1}, \dots, \mathbf{v}_{i,m_i}\} := \mathcal{N}_{k^\star}(\mathbf{b}_i)$ be the set of collected vertices, the corresponding local normality region is defined as

$$\mathcal{S}_i = \mathrm{conv}(\{\mathbf{v}_{i,1}, \dots, \mathbf{v}_{i,m_i}\})\,, \tag{7}$$

where $\mathrm{conv}(\cdot)$ denotes the convex hull and $m_i = |\mathcal{N}_{k^\star}(\mathbf{b}_i)|$. By construction, $\mathcal{S}_i$ provides an interpretable local confidence region of nominal behavior around the QAOA-driven anchors.

Example 1: Consider a simple bi-variate time series as in Fig. 1. If the QAOA optimization stage selects the sample at timestamp $t = 1$ as an anchor, yielding the point $\mathbf{b} = (1, [2.5, 1.0])$. The covering progressively expands according to (6). For $k = 1$ and $k = 2$, the neighborhood contains only normal samples, and the expansion is therefore accepted. Attempting to increase k to $k = 3$ would include a sample at timestamp $t = 4$ that is labeled as anomalous in the first channel, consequently, the expansion is halted and $k^\star = 2$. The set of collected vertices is thus given by $\mathbf{v}_1 = (0, [1.0, 2.5])$, $\mathbf{v}_2 = (2, [3.5, 1.5])$, and $\mathbf{v}_3 = (3, [1.5, 3.0])$. The corresponding local normality region is defined as the convex hull $\mathcal{S} = \mathrm{conv}(\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\})$, which is a convex confidence region of nominal behavior anchored at $\mathbf{b}$. Test samples outside $\mathcal{S}$ are predicted as anomalies.

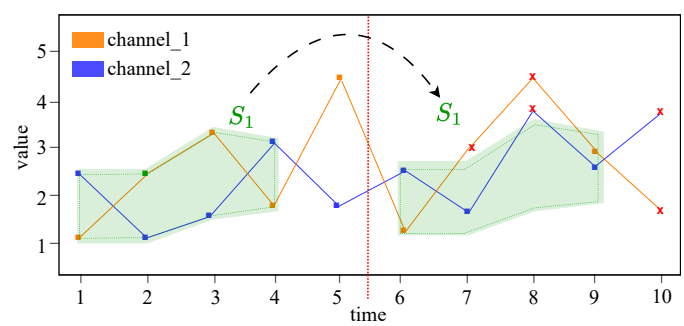

Fig. 1. Schematics of Example 1 where the resulting region obtained on the training set (left of the red dashed line) is shifted to the test set (right of the red dashed line)to detect anomalies.

Due to the hardware constraints of current NISQ devices, as the time series become more complicated, the entire pipeline is executed in a batched fashion. The training and test sets are partitioned into disjoint batches of fixed length τ, and the QAOA optimization, anchor selection, and convex-hull construction steps are independently applied within each batch (for convenience we set $k_{\max} = \tau$). The global model of nominal system behavior is obtained by aggregating all local convex regions constructed during the training phase. The complete proposed pipeline is reported in Algorithm 1.

Algorithm 1 Convex-hull Covering and Anomaly Detection

Input: Multivariate time series $\mathbf{X}$ as in (1)
Output: Set of detected anomalous timestamps A
Split $\mathbf{X}$ into training set $\mathbf{X}_{\text{train}}$ and test set $\mathbf{X}_{\text{test}}$
Partition $\mathbf{X}_{\text{train}}$, $\mathbf{X}_{\text{test}}$ into length-τ batches $B_1, \ldots, B_n$
Initialize global covering $\mathbb{S} \leftarrow \emptyset$
for all $B \in \{B_1, \ldots, B_n\}$ **do**
  Build the QUBO matrix for batch B as in Sect. II-C
  Run QAOA and obtain $\mathbf{z}$
  for all t_i such that $z_{t_i} = 1$ **do**
    Set anchor $\mathbf{b}_i \leftarrow (t_i, \mathbf{x}_{t_i})$
    Initialize $k \leftarrow 1$
    while $k \leq \tau$ **do**
      Compute $\mathcal{N}_k(\mathbf{b}_i) = \{(t, \mathbf{x}_t) \in B \mid |t - t_i| \leq k\}$
      if there is $(t, \mathbf{x}_t) \in \mathcal{N}_k(\mathbf{b}_i)$ labeled as anomalous **then**
        break
      end if
      $k \leftarrow k + 1$
    end while
    Set $k^\star \leftarrow k - 1$
    Collect vertices $\{\mathbf{v}_{i,1}, \ldots, \mathbf{v}_{i,m_i}\} \leftarrow \mathcal{N}_{k^\star}(\mathbf{b}_i)$
    Construct convex hull $\mathcal{S}_i \leftarrow \text{conv}(\{\mathbf{v}_{i,1}, \ldots, \mathbf{v}_{i,m_i}\})$
    Update covering $\mathbb{S} \leftarrow \mathbb{S} \cup \mathcal{S}_i$
  end for
end for
Initialize anomaly set $A \leftarrow \emptyset$
for all $(t, \mathbf{x}_t) \in X_{\text{test}}$ **do**
  if $(t, \mathbf{x}_t) \notin \mathbb{S}$ **then**
    $A \leftarrow A \cup \{t\}$
  end if
end for
return A

III. Experimental Evaluation

For the sake of conciseness, in the following we will illustrate the experimental evaluation of the proposed methodology only for two representative cases from the datasets introduced in Sect. II-A, namely *artificial*$_4$ and {`site name anonymized for review`}. All experiments are implemented in Python™, using standard scientific libraries together with Scikit-learn v1.6.0 and SciPy® v1.14.1. The quantum optimization stage is implemented using Qiskit® v1.3.1, with the `qiskit-aer` v0.15.0 and `qiskit-optimization` v0.6.1 modules. The codebase associated with this work, with sensitive data appropriately anonymized, is available at the companion repository {`link to repository anonymized for review`}.

A. Numerical Results

All the results presented are obtained using the architecture configuration summarized in Table II, where the circuit parameters d and ϕ were set heuristically, while the others were optimized during the training phase. Detection performance is weighed up by comparing the proposed quantum-classical approach with shallow and semi-supervised baseline methods to ensure a fair performance comparison, as fully supervised or deep learning-based architectures typically require large labeled datasets and different training assumptions, which are not aligned with the AD setting considered in this work.

We first report results obtained on the synthetic multivariate dataset *artificial*$_4$ to validate the main steps of the proposed methodology. As shown in Table III, all methods achieve very high performance, which is expected given the controlled statistical nature of the synthetic data and the clear separation between nominal and anomalous samples. In this simplified scenario, the proposed method performs comparably to classical baselines, confirming the correctness of the formulation under idealized conditions.

TABLE II
Architecture Setup

Hyperparameter	Value
QAOA depth	$p = 1$
QAOA optimizer	COBYLA
QAOA max iterations	200
d	Mahalanobis distance
ϕ	moving average (m=5)
τ, α, α'	12, -0.74, 0.26

Focusing on the results also shown in Table III for the real-world dataset {`site name anonymized for review`}, which represents a significantly more challenging scenario due to strong inter-channel correlations, non-stationary consumption patterns, and realistic noise, the proposed quantum-classical approach achieves the highest recall and F1-score among the considered methods, indicating an improved ability to detect anomalous timestamps while maintaining competitive precision and specificity. This behavior reflects the conservative nature of the covering strategy, which prioritizes anomaly containment at the cost of a limited number of false positives.

TABLE III
Average Performance Comparison

Model	Precision	Recall	F1-Score	Accuracy	Specificity
Artificial dataset: artificial$_4$					
Isolation	0.9412	0.9178	0.9293	0.9889	0.9894
LOF	0.9485	0.9284	0.9383	0.9621	0.9857
Quantum	0.9593	0.9468	0.9530	0.9784	0.9826
Real-world dataset:{`site name anonymized for review`}					
Isolation	0.8889	0.8554	0.8718	0.9167	0.9831
LOF	0.8971	0.8615	0.8789	0.8929	0.9601
Quantum	0.9029	0.9323	0.9174	0.9055	0.9167

A qualitative illustration of the detection output on the real-world dataset is reported in Fig. 2, showing three consecutive days from the test set. The figure highlights how the proposed method successfully captures both the regular daily consumption patterns (e.g., nighttime demand reductions) and localized deviations from nominal behavior. Only genuine departures from the local neighborhood density or from the global statistical model are identified as anomalous, resulting in a robust and interpretable detection outcome.

B. Semi-supervised Setting

The QUBO formulation in (2) does not rely on labeled data, as it operates exclusively on statistical deviations and multivariate similarity relationships extracted from the input time series. Nevertheless, supervision is introduced during the geometric construction of nominal regions in (7), where available anomaly labels are exploited to halt the expansion of local neighborhoods and prevent contamination of the learned

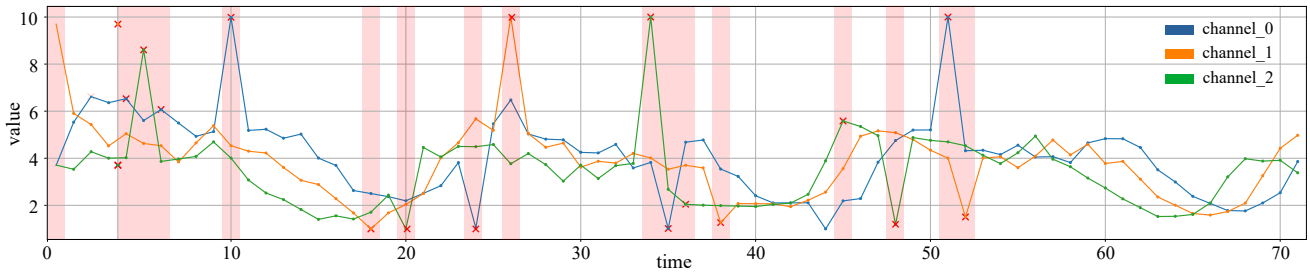

Fig. 2. Final output of the proposed detection pipeline on a portion of the test set. The multivariate time series is shown for three channels over three days. Red vertical bar sections indicate anomalous timestamps, i.e., time instants at which at least one channel falls outside the global covering region $\mathbb{S}$.

normality regions. This design reflects a realistic monitoring scenario, in which a limited amount of anomaly information is often available (like in the water distribution system analysis), while exhaustive labeling remains impractical; moreover, fully unsupervised approaches typically define global decision boundaries or density estimates that may be overly permissive in the presence of complex multivariate dependencies, whereas fully supervised or deep learning-based models require large, consistently labeled datasets that are rarely accessible [17]. In this sense, only a small fraction of anomaly labels is required, since labels are queried solely to stop expansion (cfr. Algorithm 1) rather than optimizing the QUBO.

In addition to that, unlike many semi-supervised AD approaches in which labeled samples directly influence the training objective, loss reweighting, or decision thresholds, supervision in the proposed framework does not alter the underlying optimization problem solved via QAOA [18]. Anomaly labels are introduced exclusively at the post-optimization stage, where they guide the geometric construction of nominal regions by preventing anomalous contamination during local neighborhood expansion. This separation between optimization and supervision reduces the risk of model bias and overfitting under scarce labels, while preserving the intrinsic statistical and multivariate structure encoded in the QUBO formulation. Moreover, since supervision acts directly on the geometry of the learned nominal regions, the resulting detection process remains interpretable: each detected anomaly can be explicitly explained as a violation of a convex region in the joint time–value space, rather than as the outcome of an implicit score adjustment or latent embedding shift commonly adopted by classical semi-supervised methods.

C. Discussion on the Quantum Approach

The pipeline described in Sect. II-C yields, within a single execution, a quantum state whose measurement outcomes define an empirical probability distribution over feasible binary configurations $\mathbf{z}$. Each bitstring represents a valid selection of representative samples and is associated with a probability proportional to its measurement frequency. This behavior becomes particularly relevant in structured multivariate settings, where the QUBO objective function may admit multiple global optima or a family of near-degenerate minima. Such degeneracy is not a consequence of the quantum solver, but rather an intrinsic property of the optimization problem, arising from the combinatorial nature of sample selection or from symmetries in the multivariate data geometry (a likely eventuality with real-world data), as illustrated in the following example.

Example 2: Consider the simple bivariate time series shown in Fig. 3. In this setting, the QUBO formulation, instantiated with parameters $\alpha = -0.9$ and $\alpha' = 0.1$, a global multivariate model $\phi(\cdot)$, and Mahalanobis distance for the off-diagonal interaction terms, admits two distinct global optimal configurations $\mathbf{z}_1^* = 001011110000$ and $\mathbf{z}_2^* = 000011110100$ with identical objective value. Each configuration selects a different subset of representative timestamps while being theoretically equivalent. This degeneracy is explicitly revealed by the QAOA output distribution, where both configurations appear with non-negligible measurement probability within a single execution, as illustrated by Fig. 4.

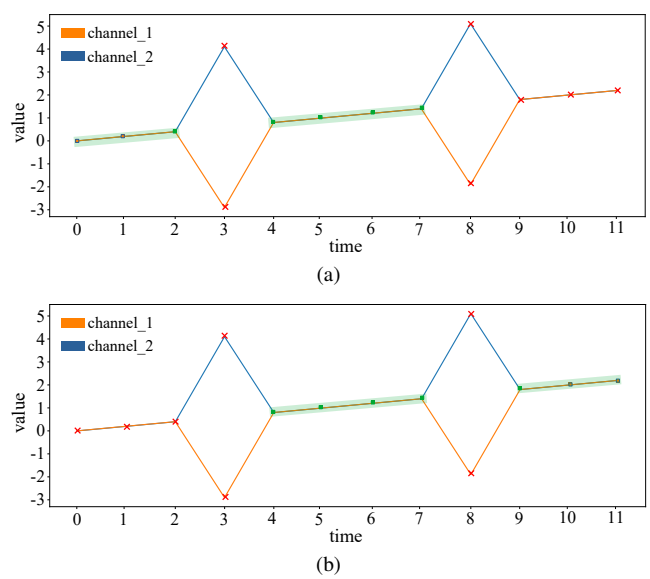

Fig. 3. Triangular events of Example 2: (a) geometric construction by optimal configuration $\mathbf{z}_1^*$, resulting in a specific selection of representative timestamps and associated convex set covering of nominal behavior; (b) alternative geometric construction by the energetically equivalent optimal configuration $\mathbf{z}_2^*$, yielding a different but valid covering of the multivariate time-value space.

A classical optimizer (e.g., `Gurobi` or `CPLEX`) typically returns a single optimal configuration per execution, implicitly selecting one solution among several energetically equivalent

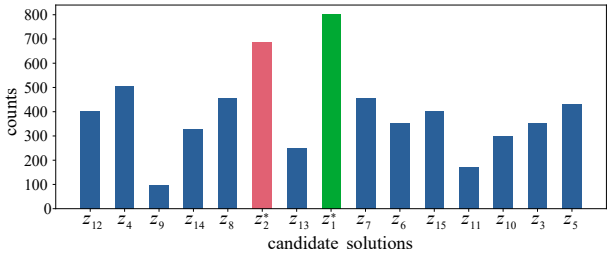

Fig. 4. Histogram of the most frequently measured bitstrings obtained from a single execution of QAOA on the bivariate dataset of Example 2. The most probable configuration ($\mathbf{z}_1^*$ in green) corresponds to one global optimum, while the second most frequent configuration ($\mathbf{z}_2^*$ in pink) represents an alternative optimal solution with identical objective value.

alternatives. The specific choice depends on solver-dependent factors such as initialization, internal heuristics, or tie-breaking rules. Recovering additional optimal configurations generally requires repeated executions with modified constraints, random perturbations, or explicit reformulations of the original optimization problem. QAOA, instead, naturally exposes the configuration that best approximates the global optimum together with additional degenerate or comparable sub-optimal solutions (configurations $\mathbf{z}^\dagger$ with non-negligible measurement counts and $f(\mathbf{z}^*) \simeq f(\mathbf{z}^\dagger)$), allowing for a direct inspection of the solution landscape without altering the objective function or retraining the model.

From an AD perspective, as mentioned in Sect. III-B, in the multivariate unsupervised/semi-supervised setting, anomalies emerge from the geometric and/or statistical interpretation of the data [19]. Contextualized to the proposed methodology, this means that each configuration $\mathbf{z}$ arising from (2) induces a set of representative samples that are used to construct nominal regions in the operative space via Algorithm 1. As also shown in the representative scenario in Fig. 3, each optimal configuration $\mathbf{z}_k$ induces a distinct geometric construction. While a systematic comparison of the detection performance associated with each configuration is beyond the scope of this work, the availability of multiple optimal solutions provides additional structural insight into the data distribution, revealing alternative but equally valid interpretations of normality that are not readily accessible through standard classical pipelines.

IV. Conclusion

In this paper, we presented a hybrid quantum–classical framework for anomaly detection in multivariate time series, extending a previously proposed univariate approach to scenarios involving multiple correlated channels. The method combines a structured multivariate QUBO formulation solved via QAOA with a set-covering strategy that leverages convex hulls, enabling robust and interpretable anomaly detection at the timestamp level, offering a different approach and partially improving the semi-supervised anomaly detection paradigm.

The proposed architecture was validated on a previously unpublished real-world dataset of water smart meter readings, demonstrating reliable detection performance in a realistic monitoring scenario and confirming the practical effectiveness of the framework in an application domain of high operational relevance. Beyond detection performance, the proposed approach provides additional capabilities that are difficult to replicate with classical solvers. In particular, the quantum optimization stage naturally exposes multiple high-quality solutions within a single training procedure, enabling alternative and complementary interpretations of normality without modifying the objective function or retraining, enabling the systematic exploration of near-degenerate configurations, that offer additional insight into the solution landscape.

References

[1] M. Gamal, A. Donkol, A. Shaban, F. Costantino, G. Di Gravio, R. Patriarca *et al.*, "Anomalies detection in smart manufacturing using machine learning and deep learning algorithms structural dynamics," in *Proceedings International Conference of Industrial Engineering and Operation Management*. IEOM Society, 2021, pp. 1611–1622.

[2] A. Yaseen, "The role of machine learning in network anomaly detection for cybersecurity," *Sage Science Review of Applied Machine Learning*, vol. 6, no. 8, pp. 16–34, 2023.

[3] A. Gaddam, T. Wilkin, and M. Angelova, "Anomaly detection models for detecting sensor faults and outliers in the iot-a survey," in *2019 13th International Conference on Sensing Technology (ICST)*. IEEE, 2019, pp. 1–6.

[4] K. Feasel, *Finding Ghosts in Your Data - Anomaly Detection Techniques with Examples in Python*. Apress, 2022.

[5] K. Shaukat, T. M. Alam, S. Luo, S. Shabbir, I. A. Hameed, J. Li, S. K. Abbas, and U. Javed, "A review of time-series anomaly detection techniques: A step to future perspectives," in *Future of Information and Communication Conference*. Springer, 2021, pp. 865–877.

[6] G. Li and J. J. Jung, "Deep learning for anomaly detection in multivariate time series: Approaches, applications, and challenges," *Information Fusion*, vol. 91, pp. 93–102, 2023.

[7] M. E. Villa-Pérez, M. A. Alvarez-Carmona, O. Loyola-González, M. A. Medina-Pérez, J. C. Velazco-Rossell, and K.-K. R. Choo, "Semi-supervised anomaly detection algorithms: A comparative summary and future research directions," *Knowledge-Based Systems*, vol. 218, p. 106878, 2021.

[8] M. Grossi, N. Ibrahim, V. Radescu, R. Loredo, K. Voigt, C. Von Altrock, and A. Rudnik, "Mixed quantum–classical method for fraud detection with quantum feature selection," *IEEE Transactions on Quantum Engineering*, vol. 3, pp. 1–12, 2022.

[9] M. Guo, S. Pan, W. Li, F. Gao, S. Qin, X. Yu, X. Zhang, and Q. Wen, "Quantum algorithm for unsupervised anomaly detection," *Physica A: Statistical Mechanics and its Applications*, vol. 625, p. 129018, 2023.

[10] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio *et al.*, "Variational quantum algorithms," *Nature Reviews Physics*, vol. 3, 2021.

[11] L. Lavagna, A. Ceschini, A. Rosato, and M. Panella, "A layerwise-multi-angle approach to fine-tuning the quantum approximate optimization algorithm," in *2024 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 2024, pp. 1–8.

[12] Blinded, "Blinded."

[13] K. Blekos *et al.*, "A review on quantum approximate optimization algorithm and its variants," *Physics Reports*, vol. 1068, pp. 1–66, 2024.

[14] F. Glover, G. Kochenberger, and Y. Du, "A tutorial on formulating and using QUBO models," *arXiv preprint arXiv:1811.11538*, 2018.

[15] E. Farhi, J. Goldstone, and S. Gutmann, "A quantum approximate optimization algorithm," *arXiv preprint arXiv:1411.4028*, 2014.

[16] D. L. Iverson, "Inductive monitoring system (ims) — a data mining tool for systems monitoring," NASA Ames Research Center, Tech. Rep. NASA/TM-2004-213028, 2004, available at https://ntrs.nasa.gov/.

[17] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[18] L. Ruff, J. Kauffmann, R. Vandermeulen, G. Montavon, W. Samek, K.-R. Müller, and M. Kloft, "A unifying review of deep and shallow anomaly detection," *Proceedings of the IEEE*, vol. 109, no. 5, pp. 756–795, 2021.

[19] A. Blázquez-García, A. Conde, U. Mori, and J. A. Lozano, "A review on outlier/anomaly detection in time series data," *ACM computing surveys (CSUR)*, vol. 54, no. 3, pp. 1–33, 2021.